

Words4You

Find your next book by how you want to FEEL

Ironhack Data Analytics · Diana Yule

1,077 books

2 sources

11 clusters

7 moods

The problem: too many books, wrong question

Genre tells you what a book IS. Mood tells you how it FEELS.

"I don't know what to read next — I just know I want something uplifting."

A reader in a bookshop, every single day.

Catalogues answer with 1,000-item lists sorted by genre.
Nobody browses 1,000 items.



Endless lists

1,077 books is already too many to scan



Genre ≠ feeling

'Fantasy' can be cozy, dark, funny or tragic

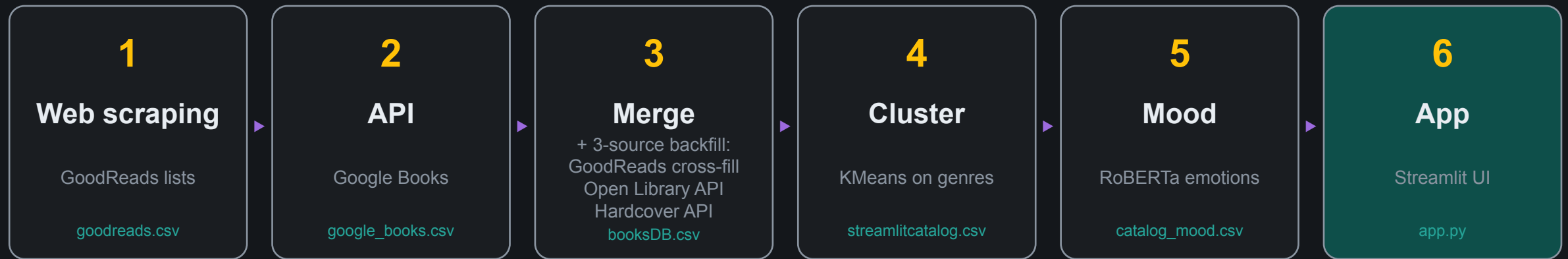


One book at a time

picked by mood, rating, length — your call

A one-way pipeline

Six stages — each notebook consumes the previous one's output. No stage ever writes backwards.



Every path lives in config.yaml — nothing hard-coded, fully re-runnable

Independent ways to get data

Two retrieval methods — we used scraping AND a REST API.

WEB SCRAPING

617

books · GoodReads Listopia lists → detail pages

requests + BeautifulSoup — parse the HTML humans see

REST API

460

books · Google Books volumes endpoint, paginated

requests — structured JSON straight from the source

Polite: random 1–6 s sleeps

Resumable: checkpoint every 25 books

None-safe: missing element → None

Merging: fill the holes

Three extra services patch missing values — match once, fill many.

GoodReads cross-fill

fuzzy title ≥ 0.90 + author ≥ 0.85 (difflib)

Enrichment 1

Open Library

Fills rating, pages, year, genres, cover

Enrichment 2

Hardcover

Matches same as ol + synopsis. Best title match ≥ 0.90

Enrichment 3

Golden rules

- Only nulls are filled — an existing value is never overwritten by a lower-trust source.
- Zero-rating gate: Hardcover returns 0.0 for unrated books — accepted only when ratings_count > 0.
- Remaining nulls are documented, not papered over.

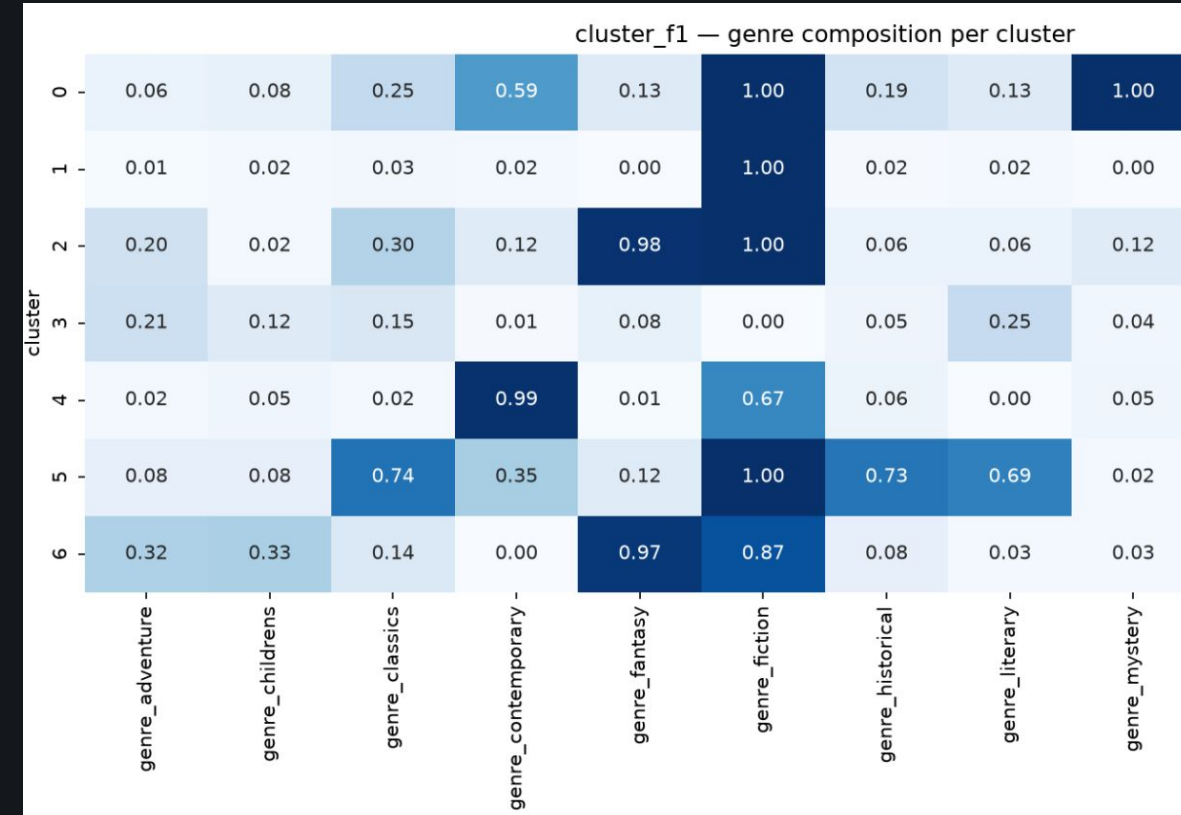
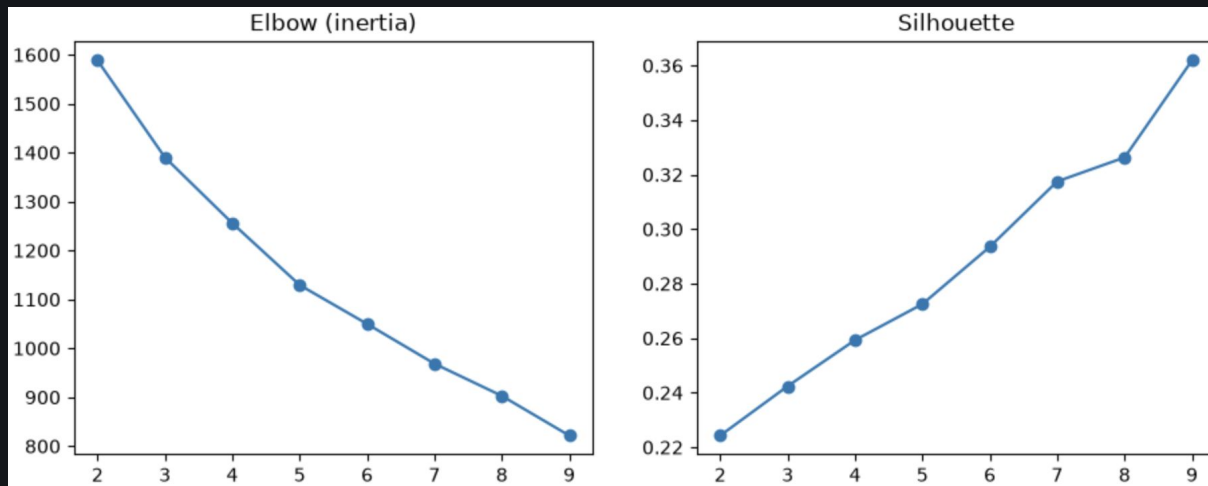
Finding the shelves: clustering

Unsupervised KMeans on a binary genre matrix.

83%

of books carry the tag 'fiction' — a stopword, not a segment. It alone defined a giant junk cluster.

Validation: genre heatmaps + min cluster size 30 — silhouette is meaningless on sparse binary data.

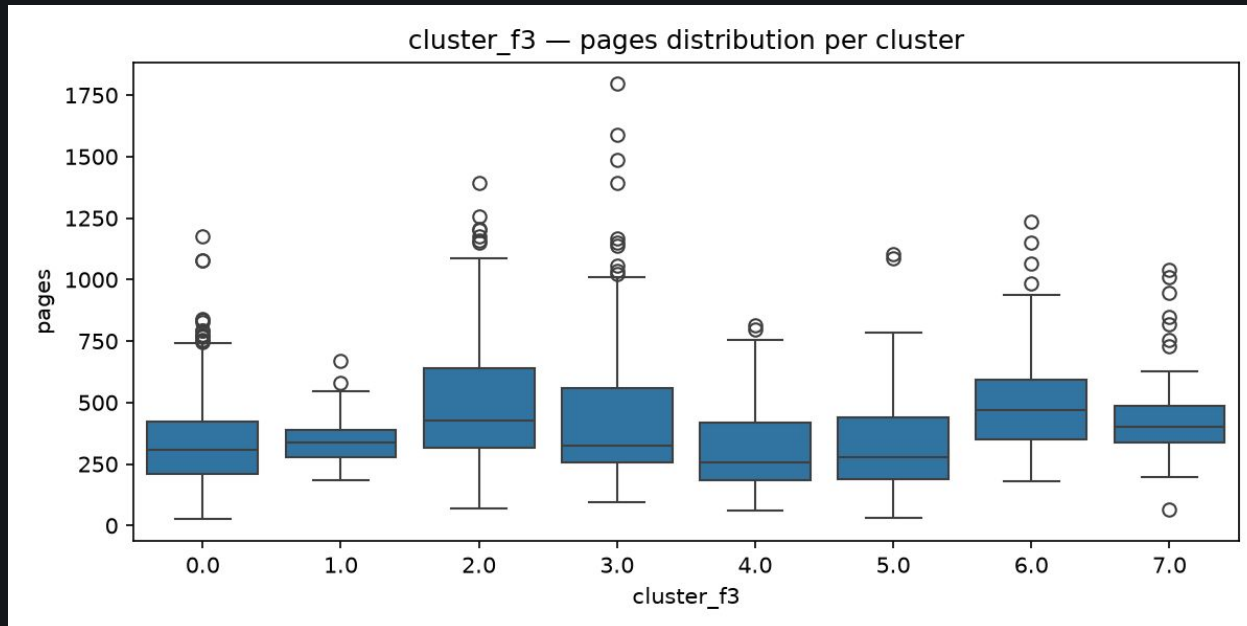
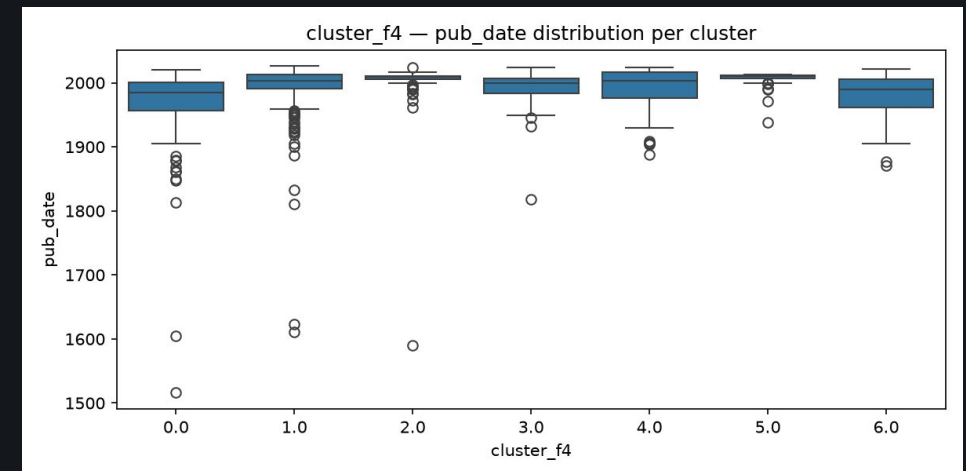
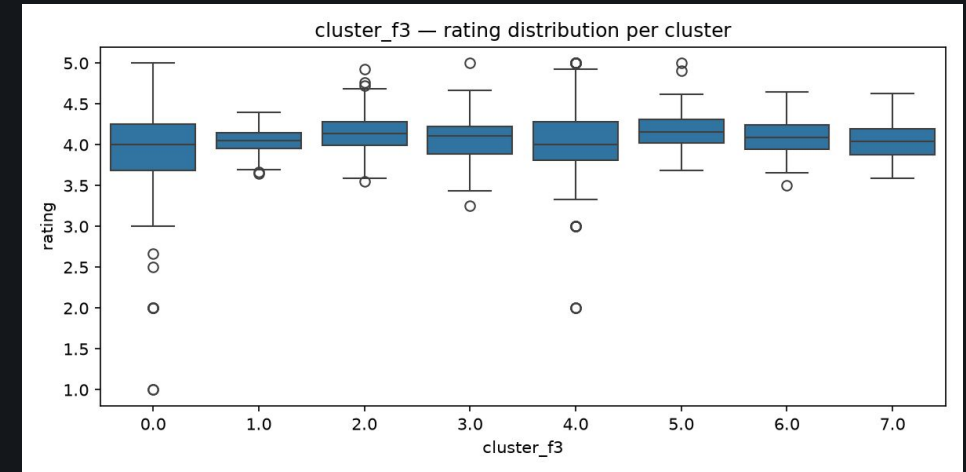


Finding the shelves: clustering

Unsupervised KMeans, checking other features combinations

No added value

Experiment was done with other features of genre + others: f2 (+pages), f3 (+rating), f4 (+year). However, they were rejected as they add more noise than improving the clusters, the app uses those features as filters.



Finding the shelves: clustering by genre

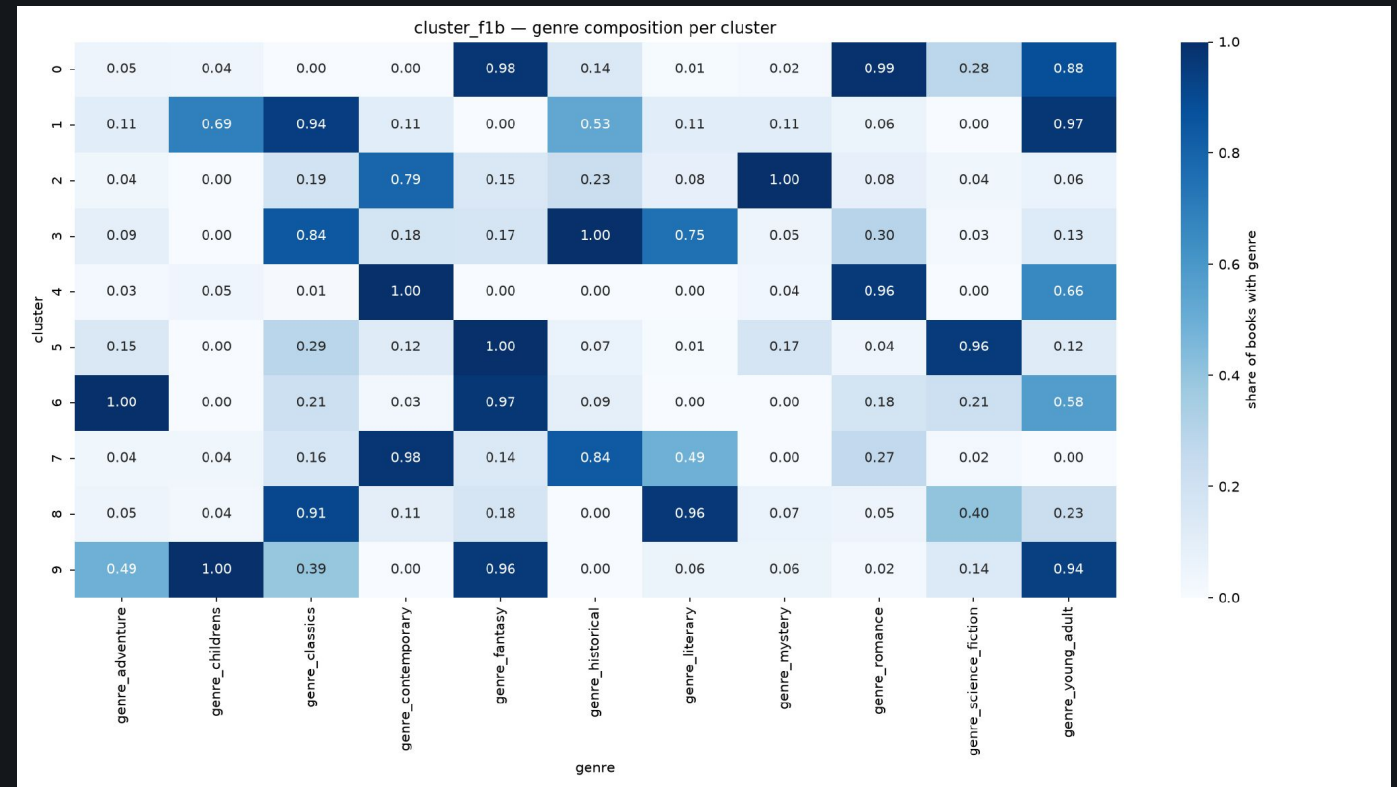
Unsupervised KMeans on a binary genre matrix — after one crucial fix.

Tag **'fiction'** — **STOPWORD**

Fix: drop it + exclude books <2 tags →
KMeans, k = 11, hand-labelled clusters like
'Romance + Fantasy'.

K values clearing min size cluster: 2 - 11

612 clusterable · 465 served via filters



Which genres define each cluster — the heatmap that picked the model

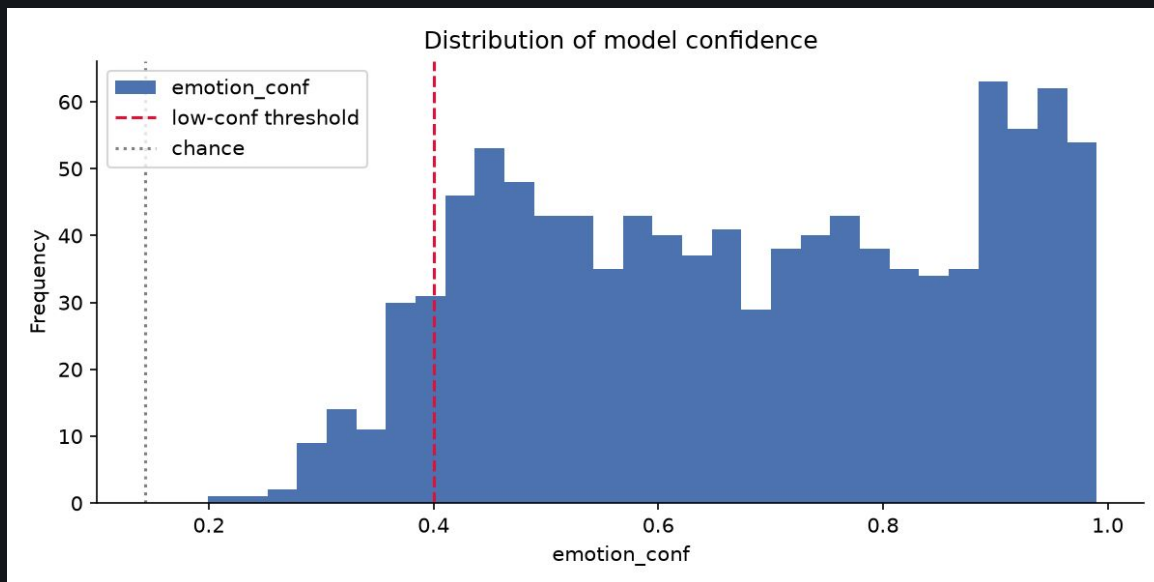
Teaching the catalogue to feel: mood

A transformer reads every synopsis and assigns one of 7 emotions — mapped to reader-facing moods.

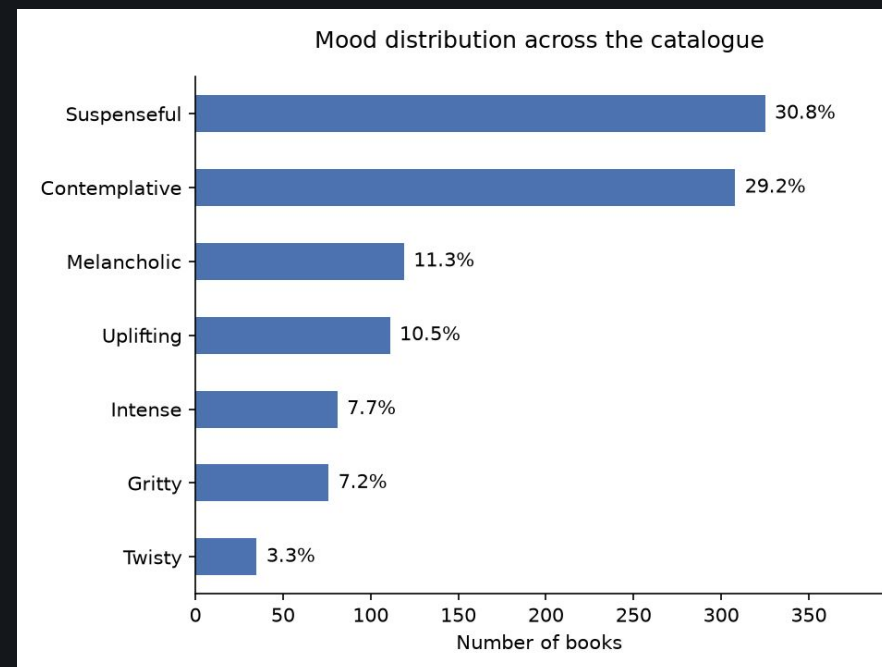
emotion-english-distilroberta-base

- 7 Ekman emotions → 7 moods:
joy (uplifting), sadness (melancholic), fear (suspenseful), anger (intense), surprise (twisty), disgust (surprise), neutral (contemplative).
- Long synopses: chunked, scored per chunk, mean-pooled — never truncated.
- 1,055 of 1,077 books scored; picks are weighted by model confidence.

Confidence check: emotion confidence (prob vector) median = 67%



Variance check



What this model can't promise

Honest limitations

Synopsis ≠ book

We classify how a book was SOLD, not what it contains. A cheerfully marketed tragedy scores as Uplifting.

Domain shift

The RoBERTA model was fine-tuned on tweets and dialogue — publisher marketing copy is different territory. The shift is real and unmeasured.

No ground truth

Nobody labelled the 'true' mood → no accuracy figure. Instead: 3 pre-registered consistency checks (median confidence 0.673 vs 0.143 chance).

90 low-confidence books (8.5%) are excluded from the mood filter

What the data taught me

Four findings that outlive this project.

One source is never enough

The interesting work was the three-source backfill, not the retrieval.

Frequent tags are stopwords

83% 'fiction' carried zero information — dropping it created real clusters.

Mood ≠ sentiment

7 emotions separate melancholic from gritty; positive/negative cannot.

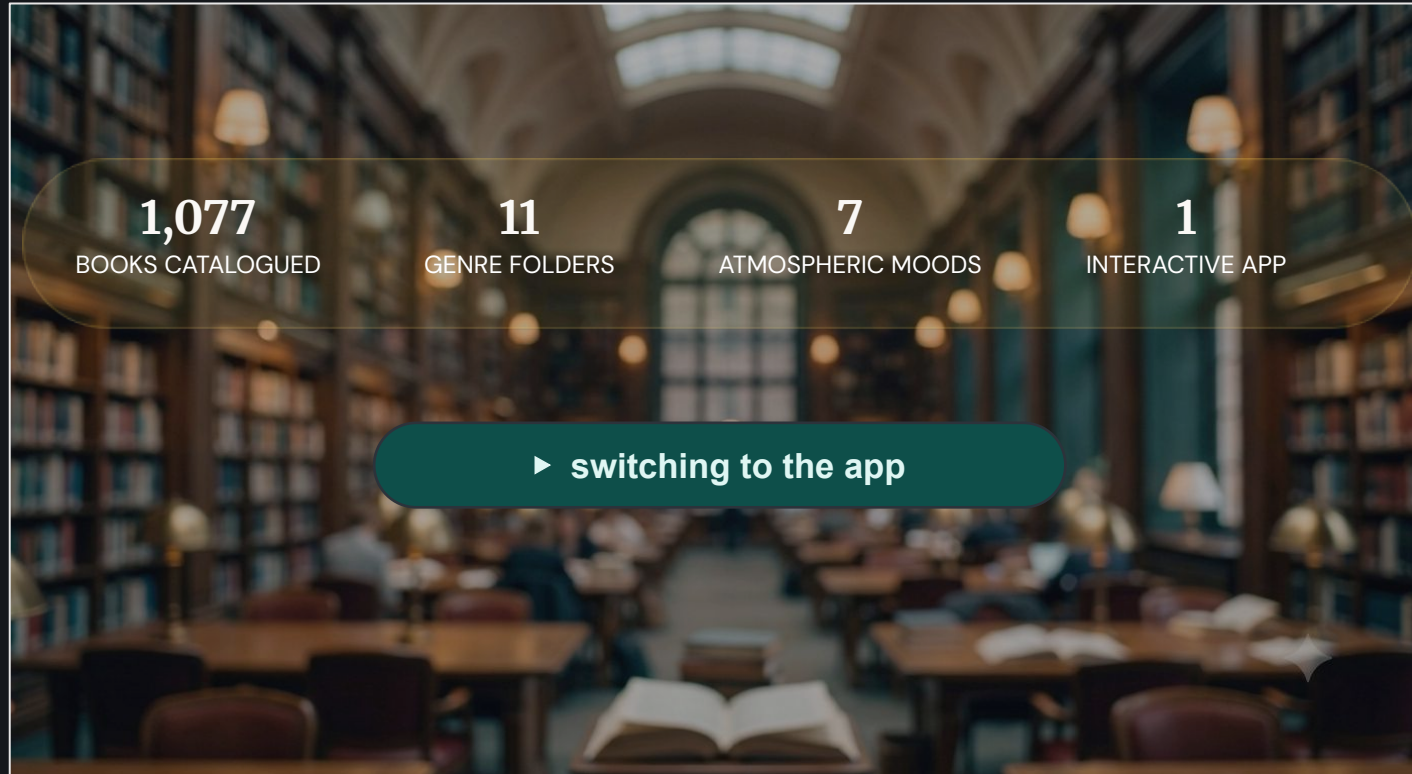
Keep user filters out

Pages, rating, year stay as filters — or the model silently overrides the user.

1,077 books · 2 sources + 3 backfills · 11 clusters · 7 moods · 1 app

Words4You — live demo

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Thank you! Any Questions?